Sankarasrinivasan Seshathiri

Computer Vision & Machine Learning Engineer · PhD

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Computer vision researcher and engineer with 10+ years across industry and academia in France, Taiwan and India. Builds deep learning systems for 6D pose estimation, object detection under occlusion, segmentation, generative imaging and virtual try-on, and has shipped fashion-AI models for e-commerce products. 30+ peer-reviewed papers (incl. IEEE TCSVT, IEEE TCE, ICIP), 2 granted patents, and reviewer for IEEE TIP, TCSVT and IJCV.

EXPERIENCE

Postdoctoral Researcher · École Centrale de Lyon, Lyon, France — May 2026 – Present

- Developing deep learning models to classify polymers from atomic force microscopy (AFM) force–distance curves.
- Building software for 3D AFM image visualisation and analysis.

Postdoctoral Researcher, Project Acrobath · LIRIS, École Centrale de Lyon, France — May 2024 – Dec 2025

- **6D pose estimation:** developed 6D pose models for known industrial forged objects and evaluated state-of-the-art models for unseen objects.
- **Detection under occlusion:** built and benchmarked detectors (DETR, YOLO and others) and integrated segmentation models to handle heavy occlusion.
- **Simulation:** ran robotic simulation studies in ManiSkill for the industrial handling use case.

Member of Technical Staff · Streamoid Technologies, Bengaluru, India — Aug 2021 – Jun 2023

Built computer vision models for three of Streamoid's AI products for fashion e-commerce.

- **Fine-grained fashion classifiers:** models for subtle clothing attributes (sleeve length, neck shape, etc.) on cropped and full-length high-resolution product images.
- **Photo-Studio:** virtual model generation pipeline combining skin and cloth segmentation, face swap, GAN-based image translation, garment warping and virtual try-on.
- **Catalogix:** hybrid segmentation model for background replacement, shadow synthesis and high-resolution catalogue images from photoshoots.
- Also worked on self-supervised learning and salient object detection for fashion imagery.

Research Assistant · National Taiwan University of Science and Technology, Taipei — Sep 2019 – Jun 2021

- AI for deep visual computing in printed-document imaging: halftone classification, watermarking, reconstruction and image quality assessment.

Teaching Assistant · National Taiwan University of Science and Technology, Taipei — Sep 2016 – Aug 2020

- Digital Image Processing (2 semesters) and Multimedia Signal Processing (3 semesters).

Junior Research Fellow · Vel Tech University, Chennai, India — Aug 2014 – Jul 2016

- Vision-based design and formation control of micro ornithopters (Indo–Taiwan project), later granted an Indian patent.
- High-speed photography processing, image processing for MAV navigation, and on-board avionics and control of flapping-wing vehicles.
- Full-field, non-contact structural health monitoring of long-span railway bridges and heritage structures (Indo–Canada project).
- Taught part-time on the M.Tech Aeronautical Engineering (UAV) programme; organised workshops on ornithopter mechanism assembly and real-time flying.

Project Engineer · Kalpataru Power Transmission, India — Feb 2010 – Jun 2011

- Field quality, project planning and scheduling; secured crossing approvals from power-line, NHAI and railway authorities.

SKILLS

Languages	Python, C++, MATLAB, Java
Deep learning & vision	PyTorch, TensorFlow, Keras, OpenCV, CUDA & GPU optimisation
3D & simulation	BlenderProc, Blender, ManiSkill
Engineering	Git & GitHub, Docker, VS Code
Expertise	6D pose estimation, object detection, segmentation, GANs & image translation, virtual try-on, self-supervised & contrastive learning, few-shot learning, knowledge distillation, image quality assessment

EDUCATION

Ph.D., Electrical Engineering · National Taiwan University of Science and Technology 2016 – 2023
Thesis: Stochastic Analysis and Hybrid Deep Learning Frameworks for Enhanced Halftone Watermarking, Classification and Reconstruction. Advisor: Prof. Jing-Ming Guo. Coursework 3.88/4.3, dissertation 4.3/4.3.

M.Tech., Electronic Systems & Communication · NIT Rourkela, India 2012 – 2014
CGPA 8.59/10. Thesis: Compressed Measurement of Structural Similarity Index.

B.Tech., Electrical & Electronics Engineering · SASTRA University, India 2005 – 2009
CGPA 8.09/10.

SELECTED PUBLICATIONS

34 peer-reviewed papers in total; full list at sankarsrin.github.io/publications

- J.-M. Guo, **S. Seshathiri**. Visually encrypted watermarking for ordered-dithered clustered-dot halftones. *IEEE Trans. Circuits and Systems for Video Technology*, 33(8), 2023.
- J.-M. Guo, A. W. H. Prayuda, H. Prasetyo, **S. Seshathiri**. Deep learning-based image retrieval with unsupervised double bit hashing. *IEEE Trans. Circuits and Systems for Video Technology*, 33(11), 2023.
- J.-M. Guo, **S. Seshathiri**, Y.-F. Liu, S.-C. Lin. Novel stacking and boundary enhanced hybrid halftone screen for laser printers. *IEEE Trans. Consumer Electronics*, 70(1), 2023.
- J.-M. Guo, J.-S. Yang, **S. Seshathiri**, H.-W. Wu. A light-weight CNN for object detection with sparse model and knowledge distillation. *Electronics*, 11(4), 2022.
- J.-M. Guo, **S. Seshathiri**. Self-supervised learning for scanned halftone classification. *IEEE ICIP*, 2023.

PATENTS

- E. Balasubramanian, **S. Sankarasrinivasan**. Vision-based control of multiple flapping-wing micro aerial vehicles and formation. Indian Patent No. 417229, 2023.
- E. Balasubramanian, **S. Sankarasrinivasan**, K. Karthik. Adaptive traffic control and mapping using unmanned aerial vehicles. Indian Patent No. 405612, 2023.

AWARDS & SCHOLARSHIPS

2016–21 Doctoral Scholarship, NTUST, Taiwan
2018 Travel Grant, APSIPA ASC 2018 (Hawaii), MOST Taiwan
2017 Travel Grant, APSIPA ASC 2017 (Kuala Lumpur), MOST Taiwan
2015 Winner, Worldwide Academic Competition, Future Idea Foundation
2014–16 Junior Research Fellowship, DST, India
2012–14 Postgraduate Scholarship, MHRD, India

SERVICE

Journal reviewer: IEEE TIP, IEEE TCSVT, IJCV, IEEE Systems Journal, Scientific Reports, Multimedia Systems, Signal Image and Video Processing, World Wide Web, JISE

Conference reviewer: IEEE ICIP (2023–26), IEEE ICME (2019), SIRS (2018)

Web Chair: IEEE ISPACS 2019, Taipei

President: Indian Cultural Association, NTUST (2018–20)